

Maximal-minor integral differences for Lipschitz perturbations

MathlibAnnex.NullLagrangian.integral_maximalMinor_fderiv_add_sub_eq_zero_of_lipschitzWith
theorem

Passes the smooth compact-perturbation identity to Lipschitz maps through local strong convergence.

Statement

Fix $n \geq 1$ and $g, u : \mathbb{R}^n \rightarrow \mathbb{R}^N$ be globally Lipschitz, with u compactly supported. Fix a set of n output indices $s = \{s_1 < \dots < s_n\}$. Define $\pi_s(y) = (y_{s_1}, \dots, y_{s_n})$ and the signed selected minor

$$M_s(f)(x) = \det(\pi_s \circ Df(x)).$$

Then

$$\int_{\mathbb{R}^n} (M_s(g+u)(x) - M_s(g)(x)) dx = 0.$$

Assumptions

There are finite constants $C_g, C_u \geq 0$ such that, for all x, y ,

$$\|g(x) - g(y)\|_\infty \leq C_g \|x - y\|_\infty, \quad \|u(x) - u(y)\|_\infty \leq C_u \|x - y\|_\infty.$$

The support convention is $\text{supp } u := \overline{\{x : u(x) \neq 0\}}$, and only this set must be compact. The fixed selection s consists of n distinct output coordinates in increasing order, so its existence entails $N \geq n$. Differentiability everywhere and a boundary condition are not assumed.

Conclusion

The same signed minor M_s occurs in both terms, and their difference is integrable with integral zero. No absolute value is placed around either determinant, and separate whole-space integrability of the two minors is not claimed.

The coordinate projection π_s is contractive for the sup norms. Postcomposition $A \mapsto \pi_s \circ A$ is a continuous linear map on operator spaces, preserves subtraction, and has norm at most one. Its determinant equals the maximal minor of the standard matrix in precisely the increasing row order.

Proof route

Smooth both maps, use one compact set for all small-parameter perturbations, pass each restricted determinant integral to the limit, and finally identify the whole-space difference.

▼ Proof steps

1. The support lemma provides a compact K containing $\text{supp } u$ and also $\text{supp } u_\varepsilon$ for every sufficiently small positive ε . Lipschitz maps are locally integrable, so mollification is linear: $(g + u)_\varepsilon = g_\varepsilon + u_\varepsilon$. These mollifications are C^∞ . Apply the smooth compact-perturbation theorem to $\pi_s \circ g_\varepsilon$ and $\pi_s \circ u_\varepsilon$; output selection preserves compact support. The chain rule identifies their derivatives with selected square operators and gives

$$\int_{\mathbb{R}^n} (M_s((g + u)_\varepsilon) - M_s(g_\varepsilon)) = 0.$$

2. Outside K , u_ε is locally zero. The two maps then agree on a neighborhood, so their derivatives, and thus their selected minors, agree there. The previous integral therefore equals its restriction to K . This argument only needs local equality; it does not require differentiability of the unsmoothed maps at every point.
3. Apply strong local convergence of mollified derivatives to $f = g + u$ and to $f = g$, on this same compact K . For $T_s(A) = \pi_s \circ A$, contractivity and linearity give

$$\|T_s(Df_\varepsilon) - T_s(Df)\|_{L^n(K)} \leq \|Df_\varepsilon - Df\|_{L^n(K)} \longrightarrow 0.$$

Continuous linear postcomposition preserves eventual L^n membership; the limiting derivative belongs to $L^n(K)$ by its Lipschitz bound. Thus all hypotheses of determinant-integral continuity hold with measure restricted to K . Its output is $\int_K M_s(f_\varepsilon) \rightarrow \int_K M_s(f)$ for each of these two maps.

4. The smooth selected minors are integrable on compact K . For the limiting maps $f = g + u, g$, the bound $|M_s(f)(x)| \leq n\|\pi_s \circ Df(x)\|_{\text{op}}^n$ and $Df \in L^n(K)$ give the same integrability. Thus both terms may be integrated separately on K . Using Step 3 for each term,

$$\begin{aligned} 0 &= \int_K (M_s((g + u)_\varepsilon) - M_s(g_\varepsilon)) \\ &= \int_K M_s((g + u)_\varepsilon) - \int_K M_s(g_\varepsilon) \\ &\longrightarrow \int_K M_s(g + u) - \int_K M_s(g) = \int_K (M_s(g + u) - M_s(g)). \end{aligned}$$

This proves the restricted difference integral is zero; it does not subtract unrestricted whole-space integrals.

5. Outside K , the original perturbation u is locally zero, so $M_s(g + u) - M_s(g) = 0$ there as well. The difference is integrable on K and equals zero outside; it is therefore integrable on the whole space and has the same integral. This proves the asserted whole-space identity without requiring mollification to preserve any boundary values.

Main citations

- Exact declaration and its source — `MathlibAnnex.NullLagrangian.integral_maximalMinor_fderiv_add_sub_eq_zero_of_lipschitz` With
- Mollification of a map — `MathlibAnnex.Mollification.mollify`

- Strong local convergence of derivatives —
MathlibAnnex.Mollification.tendsto_eLpNorm_fderiv_mollify_sub
- Smooth determinant difference identity —
MathlibAnnex.Piola.integral_det_fderiv_add_sub_eq_zero_of_contDiff
- Determinant integrals under strong convergence —
MathlibAnnex.MeasureTheory.tendsto_integral_det_of_strongLn
- Increasing output-coordinate selection as a linear map —
MathlibAnnex.ContinuousLinearMap.selectedOutputLinearMap
- Contractivity of coordinate selection —
MathlibAnnex.ContinuousLinearMap.norm_selectedOutputLinearMap_le
- Continuous linear output selection — MathlibAnnex.ContinuousLinearMap.selectedOutput
- Evaluation at the selected increasing coordinate —
MathlibAnnex.ContinuousLinearMap.selectedOutput_apply
- Selected square operator by postcomposition —
MathlibAnnex.ContinuousLinearMap.selectedSquare
- Postcomposition on operator spaces — MathlibAnnex.ContinuousLinearMap.selectedSquareCLM
- Evaluation of the operator-space selection map —
MathlibAnnex.ContinuousLinearMap.selectedSquareCLM_apply
- Operator norm of output selection at most one —
MathlibAnnex.ContinuousLinearMap.norm_selectedOutput_le_one
- Contractivity of selected square operators —
MathlibAnnex.ContinuousLinearMap.norm_selectedSquare_le
- Selection commutes with subtraction — MathlibAnnex.ContinuousLinearMap.selectedSquare_sub
- Selected determinant equals the signed increasing-row minor —
MathlibAnnex.ContinuousLinearMap.det_selectedSquare
- Signed selected minor of a derivative — MathlibAnnex.NullLagrangian.maximalMinorIntegrand
- Compact support under output selection —
MathlibAnnex.NullLagrangian.hasCompactSupport_selectedOutput
- Smooth selected-minor difference identity —
MathlibAnnex.NullLagrangian.integral_maximalMinor_fderiv_add_sub_eq_zero_of_contDiff
- Continuity of a mollified selected minor —
MathlibAnnex.NullLagrangian.continuous_maximalMinorIntegrand_mollify
- Convergence of selected-minor integrals on a compact set —
MathlibAnnex.NullLagrangian.tendsto_integral_maximalMinor_mollify
- Integrability of Lipschitz minors on a compact set —
MathlibAnnex.NullLagrangian.integrableOn_maximalMinor_fderiv_of_lipschitzWith
- The minor difference vanishes off the perturbation support —
MathlibAnnex.NullLagrangian.topMinor_difference_eq_zero_of_not_mem_tsupport
- Restricting the minor difference integral to its support —
MathlibAnnex.NullLagrangian.integral_topMinor_difference_eq_setIntegral

Lean source signature (exact)

```
theorem integral_maximalMinor_fderiv_add_sub_eq_zero_of_lipschitzWith
  {m N : ℕ} (s : Matrix.MaximalMinorIndex (m + 1) (Fin N))
  {g u : (Fin (m + 1) → ℝ) → (Fin N → ℝ)}
  {Cg Cu : ℝ≥0} (hg : LipschitzWith Cg g) (hu : LipschitzWith Cu u) (huc :
HasCompactSupport u) :
```

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    ∫ x, (maximalMinorIntegrand s (fun y => g y + u y) x -
        maximalMinorIntegrand s g x) = 0

```

In the source	Mathematical meaning
<pre> {m N : ℕ} (s : Matrix.MaximalMinorIndex (m + 1) (Fin N)) </pre>	The domain dimension $n = m + 1 \geq 1$ and fixed increasingly ordered selection $s = \{s_1 < \dots < s_n\}$ of n output coordinates.
<pre> {g u : (Fin (m + 1) → ℝ) → (Fin N → ℝ)} </pre>	The base map g and perturbation u from $\mathbb{R}^n$ to $\mathbb{R}^N$, with sup norms.
<pre> {Cg Cu : ℝ≥0} (hg : LipschitzWith Cg g) (hu : LipschitzWith Cu u) </pre>	The global bounds $\ g(x) - g(y)\ _\infty \leq C_g \ x - y\ _\infty$ and $\ u(x) - u(y)\ _\infty \leq C_u \ x - y\ _\infty$ for all x, y , with nonnegative constants.
<pre> (huc : HasCompactSupport u) </pre>	The perturbation has compact support $\overline{\{x : u(x) \neq 0\}}$.

In the source	Mathematical meaning
<pre> maximalMinorIntegrand s (fun y => g y + u y) x - maximalMinorIntegrand s g x </pre>	The signed difference $M_s(g + u)(x) - M_s(g)(x)$, where $M_s(f)(x) = \det(\pi_s \circ Df(x))$ and $\pi_s(y) = (y_{s_1}, \dots, y_{s_n})$.
<pre> ∫ x, (maximalMinorIntegrand s (fun y => g y + u y) x - maximalMinorIntegrand s g x) = 0 </pre>	The whole-space Lebesgue integral of this difference equals zero. The same increasing selection s occurs in both terms, without absolute values.

▼ Exact surrounding binder context (separate excerpt)

```

noncomputable section
open Set MeasureTheory Filter
open scoped BigOperators Topology ENNReal NNReal
namespace MathlibAnnex
namespace NullLagrangian

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▼ Exact content identity

Declaration: `MathlibAnnex.NullLagrangian.integral_maximalMinor_fderiv_add_sub_eq_zero_of_lipschitzWith`

Accepted content SHA-256: 633190ad10fc198b21cc65efb19b0cb7005ec47f94d93528baa9aa946cfd1d58

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